

PROMYS INDIA -2023

Q.1 Consider the sequence

$$a_1 = 2^1 - 3 = -1,$$

$$a_2 = 2^2 - 3 = 1,$$

$$a_3 = 2^3 - 3 = 5,$$

$$a_4 = 2^4 - 3 = 13,$$

$$\cdot$$

$$a_n = 2^n - 3,$$

.

defined for positive integers n . Which elements of this sequence are divisible by 5? What about 13? Are any elements of this sequence divisible by $65 = 5 \cdot 13$? Why or why not?

Solution: -

Every fourth number that occurs after 5 in this sequence is divisible by 5. Every twelfth number that occurs in this sequence is divisible by 13. According to my conjecture, there is no such number divisible by 65. There is only one element divisible by 65, that is " ∞ ".

Let's take an example of a small number '15'

$15 = 5 \times 3$, here numbers which are not divisible by either 5 or 3 will be denoted by "." and numbers which are divisible by either 5 or 3 will be denoted by "-".

	3	5	→ product of prime numbers
1.	.	.	
2.	.	.	
3.	(-)	.	→ here, if we observe, '-' of 3 misses a chance to get equal with '-' of 5, by 2 numbers, we will denote this by "+2".
4.	.	.	
5.	.	(-)	
6.	-	.	
7.	.	.	
8.	.	.	
9.	(-)	.	→ here, it misses by 1, so "+1".
10.	.	(-)	
11.	.	.	
12.	-	.	
13.	.	.	

14. . .
15. $\ominus$ $\ominus$ $\longrightarrow$ at this point, both 3 and 5 are divisible,
so 15 is also divisible. (15 = 5 x 3) “+0”

By observing the multiples of 15, we can conclude that it follows a decreasing order pattern, ‘+2>+1>+0’ or “**2>1>0**”. For any number with product of its prime numbers follow this pattern.

Getting back to our question, is there any number divisible by 65 in this sequence?

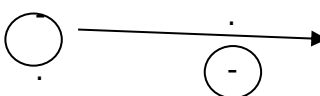
In case of 15, we observed that both 3 and 5 started with a common number followed by its sequence, both started with 1, followed by +1 sequence or number line sequence.

$$a_n = 2^n - 3$$

First number divisible by 5 is “a₃” and first number divisible by 13 is “a₄”. Here, numbers have a gap, to fix this we will start with a point where both numbers share the same number and not divisible by 5 and 13, “a₅”.

- | | | | | |
|-----|--------------|-----------|-----------|---|
| 1. | $\ominus$ 29 | . | . | $\longrightarrow$ we are starting with 29 as it |
| 2. | 61 | . | . | is not divisible by either 5 or |
| 3. | 125 | - | . | 13 and also stands for “a ₅ ”. |
| 4. | 253 | . | . | |
| 5. | 509 | . | . | |
| 6. | 1021 | . | . | |
| 7. | 2045 | - | . | |
| 8. | 4093 | . | . | |
| 9. | 8189 | . | . | |
| 10. | 16381 | . | . | |
| 11. | 32765 | $\ominus$ | . | $\longrightarrow$ here, ‘-’ of 5, misses by 1, |
| 12. | 65533 | . | $\ominus$ | so will write it as “+1”. |
| 13. | 131069 | . | . | |
| 14. | 262141 | . | . | |
| 15. | 524285 | - | . | |
| 16. | 1048573 | . | . | |
| 17. | 2097149 | . | . | |
| 18. | 4194301 | . | . | |
| 19. | 8388605 | - | . | |
| 20. | 16777213 | . | . | |
| 21. | 33554429 | . | . | |
| 22. | 67108861 | . | . | |

23. 134217725
 24. 268435453



here, again it misses by 1, so will write "+1".

If we notice, at **11**. '-' of 5, misses by 1 or "+1". Again, this happens at **23**. misses by 1 or "+1". If we write and practice this sequence till " ∞ ", they will continue missing the numbers by 1 as it forms a loop of "+1". If this phenomenon continues, we can conclude that "-" of 5 and "-" of 13 will never occur in a same line. By using this statement, we can convey that there are no numbers divisible by 65 or **13 x 5** in $a_n = 2^n - 3$.

We can also start with a_3 to get more proofs regarding this conjecture. ("**+1**" loop)

Q.2 To get the *echo* of a positive integer, we write it twice in a row without a space. For example, the echo of 2023 is 20232023. Is there a positive integer whose echo is a perfect square? If so, how many such positive integers can you find? If not, explain why not.

Solution: -

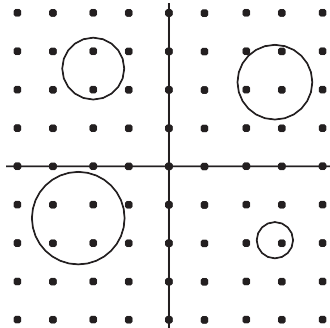
There is no positive integer whose echo is a perfect square. To prove this let's see my conjecture.:-

		<u>Squares</u>	<u>Echo</u>	<u>Difference</u>
1 x 1	=	1	11	10
2 x 2	=	4	22	18
3 x 3	=	9	33	24
4 x 4	=	16	44	28
5 x 5	=	25	55	30
6 x 6	=	36	66	30
7 x 7	=	49	77	28
8 x 8	=	64	88	24
9 x 9	=	81	99	18

Here, if we observe at 5^2 and 6^2 both the numbers share the exact difference. Following the order, 18 starts from 2^2 and ends again at 9^2 . By following this order, we can say that there is **no positive integer whose echo is a perfect square**. For 99 numbers (highest two-digit number), the difference increasing-decreasing pattern continues. For 999 numbers, the pattern continues. This pattern forms a loop of increasing-decreasing pattern. For n numbers the pattern follows.

Let's assume that patterns tends to fall and keeps on decreasing till 0. Once, a number would occur whose difference between echo and square is 0 but, as we know the pattern never falls to 0. So we can conclude that there is no number whose "**echo = square**".

Q.3 A lattice point is a point (x, y) in the plane, both of whose coordinates are integers. It is easy to see that every lattice point can be surrounded by a small circle which excludes all other lattice points from its interior. It is not much harder to see that it is possible to draw a circle which has exactly two lattice points in its interior, or exactly 3, or exactly 4, as shown in the picture below.



Do you think that for every positive integer n there is a circle in the plane which contains exactly n lattice points in its interior? Justify your answer.

Solution:-

Yes, for every positive integer n there is a circle in the plane which contains exactly n lattice points in its interior. Instead of proving this, let's rephrase the question as "Is there a specific point from which concentric circles could be made, in such a way that it contains exactly n lattice points in its interior".

Let's convert our plane into cartesian plane, so that we get close to our magic number.

$x = (\sqrt{5}, 1.1)$, this number has an outstanding property, that will solve our question.

From this point, distance from all different lattice points will be different. This states that there won't be two lattice points with same distance. Also, if we started expanding our concentric circle it won't hit two points at a time.

By using measure theory, we can prove that almost all lattice points have magic property like $(\sqrt{5}, 1.1)$.

There are countable number of lattice points in our plane. Also, there are countable number of pairs in this plane. For any two pairs of lattice points, there is an equidistant line passing through.

If a point lies on this equidistant line, then that's a bad point. If there are countable number of pairs and there would be countable number of lines.

Each line in this plane has measure 0, a countable number of sets measure 0 then the entire plane does not have measure zero now, that's infinite measure.

That means that these lines cover a set of measure 0 in entire plane, which states that almost all points on this plane are not on this equidistant line or **all points have that property** that of $(\sqrt{5}, 1.1)$.

Q.4 Calculate each of the following:

$$\begin{aligned}1^3 + 5^3 + 3^3 &= ?? \\16^3 + 50^3 + 33^3 &= ?? \\166^3 + 500^3 + 333^3 &= ?? \\1666^3 + 5000^3 + 3333^3 &= ?? \\... &= ??\end{aligned}$$

What do you see? Can you state and prove a generalization of your observations?

Solution: -

$$\begin{aligned}1^3 + 5^3 + 3^3 &= 153 \\16^3 + 50^3 + 33^3 &= 165033 \\166^3 + 500^3 + 333^3 &= 166500333 \\1666^3 + 5000^3 + 3333^3 &= 166650003333\end{aligned}$$

$$\begin{aligned}a^3 + b^3 + c^3 - 3abc &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \\a^3 + b^3 + c^3 &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) + 3abc\end{aligned}$$

Let's use our identity to get a brief about our question,

$$\begin{aligned}1^3 + 5^3 + 3^3 &= (1+5+3)(1 + 25 + 9 - 5 - 3 - 15) + 45 \\&= (9)(12) + 45 \\&= (9)(12) + (9)(5) \\&= (9)(12 + 7) \text{ or } \mathbf{(9)(17)} \\ \text{For } 16^3 + 50^3 + 33^3 &= (99)(867) + 79200 \text{ or } \mathbf{(99)(1667)}\end{aligned}$$

Let's reframe this question:

$$\begin{aligned}\mathbf{153} &= \mathbf{9 \times 17} \\ \mathbf{165033} &= \mathbf{99 \times 1667} \\ \mathbf{166500333} &= \mathbf{999 \times 166667}\end{aligned}$$

Pattern 1: after every sequence, another 9 gets added or $9 \times 10^n + (\text{previous seq.})$

$$\begin{aligned}\text{eg- } 9 \times 10^0 &= x \\ 9 \times 10^1 + (x \text{ or } 9 \times 10^0) &= x \\ 9 \times 10^2 + (x \text{ or } 99) &\dots\end{aligned}$$

Pattern 2: here, between 1 and 7, after every sequence even number of 6's gets added.

$$\begin{aligned}\text{eg- } 1 \ 7 \ (0) \\ 1 \ 6 \ 6 \ 7 \ (2) \\ 1 \ 6 \ 6 \ 6 \ 6 \ 7 \ (4) \dots\end{aligned}$$

$$\mathbf{a^3 + b^3 + c^3 = \text{pattern1} \times \text{pattern2}}$$

- Q5.** Any positive integer can be written in binary (also called base 2). For example, 37 is 100101 in binary (because $37 = 2^5 + 2^2 + 2^0$), and 45 is 101101 in binary. Let's say that a positive integer is "scattered" if, in its binary expansion, there are never two ones immediately next to each other. For example, 37 is scattered but 45 is not. How many scattered numbers are there less than 4? Less than 8? Less than 2^n ?

Solution: -

Scattered numbers = never two ones immediately next to each other in its binary expansion.

First, we will write all binary numbers from 1 to 16, to understand the pattern.

1.	1	
2.	10	(2^1)
3.	11	
4.	100	(2^2)
5.	101	
6.	110	
7.	111	
8.	1000	(2^3)
9.	1001	
10.	1010	
11.	1011	
12.	1100	
13.	1101	
14.	1110	
15.	1111	
16.	10000	(2^4)

Scattered numbers less than 4, 2

Scattered numbers less than 8, 4

Scattered numbers less than 16, 7

Scattered numbers less than 32, 12

Scattered numbers less than 64, 20...

Scattered numbers less than 2^n ,

2	4	7	12	20
2^2	2^3	2^4	2^5	2^6

If we observe our table,

To get number of scattered numbers under 2^6 or 64, we need to add 7 (number of scattered numbers in 2^4) and 12 (number of scattered numbers in 2^5).

$(7 + 12) + 1 = 20$.

By this we conclude that,

To get number of scattered numbers in 2^n ,

x = number of scattered numbers in 2^{n-1}

y = number of scattered numbers in 2^{n-2}

sum of x and y +1.

- Q.6** The set S contains some real numbers, according to the following three rules.

(i) $\frac{1}{1}$ is in S .

(ii) If $\frac{a}{b}$ is in S , where $\frac{a}{b}$ is written in lowest terms (that is, a and b have highest common factor 1), then $\frac{b}{2a}$ is in S .

(iii) If $\frac{a}{b}$ and $\frac{c}{d}$ are in S , where they are written in lowest terms, then $\frac{a+c}{b+d}$ is in S .

d

$b+d$

These rules are exhaustive: if these rules do not imply that a number is in S , then that number is not in S . Can you describe which numbers are in S ?

Solution: -

To describe numbers in S , we have to go through some calculations.

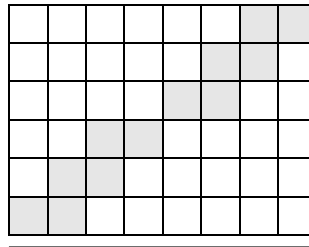
0/1, as mentioned in second rule, both numbers are written in their lowest forms, then $\frac{b}{2a}$ is in S . $0 = a$, whereas $1 = b$ but, after the conversion it turns to $\frac{1}{2 \times 0}$ or $\frac{1}{0}$. $\frac{1}{0}$ is not defined in mathematics. So, all numbers should be positive and should have their HCF as 1 or no factors in common (co-prime numbers).

By using the above calculation,

$\frac{x}{y}$ is in S if

- (i) x and y positive integers.
- (ii) HCF of x and y is 1. (**co-prime numbers**)

- Q.7** The rectangular floor of a bathroom is covered with square tiles (all of the same size). A spider starts at one corner of the bathroom, and walks to the corner diagonally opposite. For example, the figure below shows a 6×8 bathroom, in which the spider touches 12 tiles on its path. (A spider doesn't touch a tile if it just walks over the grout at the corner of a tile.) For an m by n bathroom, how many tiles does the spider touch on its walk?



Solution: -

For a rectangle of dimensions 6×8 , on which the spider touches 12 tiles on its path. Let's take another rectangle of dimensions 3×4 , here the output is 6.

Another rectangle of dimension 3×8 , here the output is 10.

Now we will convert this data into a table to get a brief overview about this question.

<u>Dimensions</u>		<u>Product</u>	<u>Addition</u>	<u>Result</u>
6×8	=	48	14	12
3×4	=	12	7	6
3×8	=	24	11	10

As addition and result are very close to each other, we will observe their relation.

$14 - 12$	=	2
$7 - 6$	=	1
$11 - 10$	=	1

—————> the result here misses by 2,1,1 respectively.

(Difference of addition and result)

Formula for $m \times n$:

$\text{addition}(m + n) - 2 = \text{result}(6, 8)$
 $\text{addition}(m + n) - 1 = \text{result}(3, 4)$
 $\text{addition}(m + n) - 1 = \text{result}(3, 4)$

2,1,1 are following an unknown property between m and n.

$\left(\begin{array}{l} 6, 8 = 2 \\ 3, 4 = 1 \\ 10, 11 = 1 \end{array} \right) \rightarrow$ now, lets rephrase the question.
 Which property does this set following?

Here, 2 is H.C.F of 6, 8. 1 is highest common factor between 3,4 and 10,11.
 By this we can conclude the formula as:
 $(m + n) - \text{HCF}(m, n) = \text{result}$

Q.8 Let P_0 be an equilateral triangle of area 10. Each side of P_0 is trisected into three segments of equal length, and the corners of P_0 are snipped off, creating a new polygon (in fact, a hexagon) P_1 . What is the area of P_1 ? Now repeat the process to P_1 – i.e. trisect each side and snip off the corners – to obtain a new polygon P_2 . What is the area of P_2 ? Now repeat this process infinitely to create an object P_∞ . What can you say about the shape P_∞ ? What is the area of P_∞ ?

Solution:-

In figure P_0 , there are 3 regular sides forming an equilateral triangle. After trisecting into three segments of equal length to make new polygon P_1 with 6 regular sides. The area of P_1 is 6.667 unit².

Area of regular polygon
 with side length “s” and $= \frac{s^2 n}{4 \tan[\frac{180}{n}]}$
 number of sides “n”

Let's take wheel as a perfect square rolling on surface, if we observe this case “the wheel is in contact with the surface” as the wheel is perfect circle and the surface here is perfectly flat. The point of contact between the wheel and surface is infinite decimals small. Can we consider **the point of contact as a side of circle?**

If we roll a circle 1 million times, it would never stop at the same point, by this we can finalize that circle has ∞ **sides**.

P_∞ will have a shape similar to a ‘**perfect circle**’, as the number of sides keep on multiplying from 3 to 6, 6 to 12, till ∞ .

Area of $P_\infty =$

$$\frac{\infty^2 n}{4 \tan \left[\frac{180}{n} \right]}$$

$$\frac{\infty}{4 \tan \left[\frac{180}{\infty} \right]}$$

$$\frac{\infty}{4 \tan(0)}$$

$$\frac{\infty}{4 \times \frac{0}{1}}$$

$$\frac{\infty}{\left(\frac{0}{1}\right)}$$

$$\infty \times \frac{1}{0}$$

$$\frac{\infty}{0}$$

Infinity divided by any number (0) is infinity.

Area of P_{∞} = " ∞ "

- Q.9** A monkey has filled in a 3×3 grid with the numbers 1, 2, . . . , 9. A cat writes down the three numbers obtained by multiplying the numbers in each horizontal row. A dog writes down the three numbers obtained by multiplying the numbers in each vertical column. Can the monkey fill in the grid in such a way that the cat and dog obtain the same lists of three numbers? What if the monkey writes the numbers 1, 2, . . . , 25 in a 5×5 grid? Or 1, 2, . . . , 121 in a 11×11 grid? Can you find any conditions on n that guarantee that it is possible or any conditions that guarantee that it is impossible for the monkey to write the numbers 1, 2, . . . , n^2 in an $n \times n$ grid so that the cat and the dog obtain the same lists of numbers?

Solution: -

Yes, monkey can fill the grid in such a way that cats and dogs obtain the same list of three numbers, if required in any order.

5	3	2
6	7	4
1	8	9

The dog here could read 30,164,72 but, cat could the same as 30,72,164.

I guarantee that it is not possible for monkey to write the numbers from 1,2,... n^2 in $n \times n$, grid so that the cat and dog obtain the same list of numbers.

For an $n \times n$ grid, take the prime numbers greater than $n^2/2$ and less than n^2 . Here, $\frac{(3)(3)}{2}$ and 3^2 , 4.5 and 9.

Here, 5 and 7 are only two prime numbers,
= 35

$$\begin{aligned} 5 \times 7 \\ 5 \times y = 35, \end{aligned}$$

here no other number than 7 could satisfy this equation. So, prime numbers should not be written in x axis or y axis.

So, if number of primes in the range n^2/n to n^2 is greater than n , then its **not possible** to solve, with the same order (correct order).

Q.10 On a strange railway line, there is just one infinitely long track, so overtaking is impossible. Any time a train catches up to the one in front of it, they link up to form a single train moving at the speed of the slower train. At first, there are three equally spaced trains, each moving at a different speed. You watch, and eventually (after all the linking that will happen has happened), you count the trains. You wonder what would have happened if the trains had started in a different order (but each of the original three trains had kept its same starting speed). On average (averaging over all possible orderings), how many trains will there be after a long time has elapsed? What if at the start there are 4 trains (all moving at different speeds)? Or 5? Or n ?

Solution: -

Here let's number of trains be 3. Imagine three trains moving left to right with different speeds. Again, think of these three trains in all possible ordering,

<u>Possible orderings</u>			<u>Number of trains</u>
1	2	3	3 (No clashes)
1	3	2	2
2	3	1	1
2	1	3	2
3	2	1	1
3	1	2	2

$$\text{Average} = \frac{\text{Number of trains}}{\text{Number of formations}} = \frac{3+1+1+2+2+2}{6} = \frac{11}{6}$$

Now we will observe the answer,

$$\frac{3 + 1 + 1 + 2 + 2 + 2}{6}$$

$$\frac{6}{6} + \frac{3}{6} + \frac{2}{6}$$

$$1 + \frac{1}{2} + \frac{1}{3}$$

If another train is added to the set,

<u>Possible orderings</u>				<u>Number of trains</u>
1	2	3	4	4

1	2	4	3	3
1	3	2	4	3
1	3	4	2	3
1	4	2	3	2
1	4	3	2	2
2	1	3	4	3
2	1	4	3	2
2	3	1	4	3
2	3	4	1	3
2	4	1	3	2
2	4	3	1	2
3	1	2	4	2
3	1	4	2	2
3	2	1	4	2
3	2	4	1	2
3	4	1	2	2
3	4	2	1	2
4	1	2	3	1
4	1	3	2	1
4	2	1	3	1
4	2	3	1	1
4	3	1	2	1
4	3	2	1	1

$$\text{Average} = \frac{\text{Number of Trains}}{\text{Number of Formations}} = \frac{1+1+1+1+1+1+2+2+2+2+2+2+2+2+2+2+3+3+3+3+3+3+4}{24}$$

$$\frac{50}{24} \text{ or } \frac{25}{12}.$$

Let's observe our answer one more time,

$$\frac{12 + 6 + 4 + 3}{12}$$

$$\frac{12}{12} + \frac{6}{12} + \frac{4}{12} + \frac{3}{12}$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

Our answer follows a pattern, in the previous question with 3 trains it showed us the same result as this without $\frac{1}{4}$. With increase in number of trains by 1, the answer added $1/n$.

If we observe this order, the **formula for n number of trains is**

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

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